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Tracking Body Composition Over a Competitive Season in Elite Soccer Players Using Laboratory- and Field-Based Assessment Methods

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Abstract

Bongiovanni, T, Lacome, M, Rodriguez, C, and Tinsley, GM. Tracking body composition over a competitive season in elite soccer players using laboratory- and field-based assessment methods. *J Strength Cond Res* 38(3): e104–e115, 2024—The purpose of this study was to describe body composition changes in professional soccer players over the course of a competitive playing season and compare the ability of different assessment methods to detect changes. Twenty-one elite male soccer players (age: 23.7 ± 4.8 years; height: 185.0 ± 5.2 cm; body mass: 80.7 ± 5.5 kg; body fat: 12.8 ± 2.2%) playing for an Italian national second league (Serie B) championship team were assessed at 4 time points throughout a competitive season: T0 (mid-October), T1 (mid-December), T2 (mid-February), and T3 (end of April). Dual-energy x-ray absorptiometry (DXA), skinfolds (SKF), and bioelectrical impedance analysis were performed at each time point, and multiple SKF-based equations were applied. A modified 4-compartment (4C) model was also produced. Data were analyzed using repeated measures analysis of variance, relevant post hoc tests, and Pearson's correlations. Dual-energy x-ray absorptiometry, 4C, and the SKF-based equations of Reilly and Civar detected differences in fat-free mass (FFM) between time points, with the most differences observed for DXA. Fat-free mass increased from T0 values to a peak at T2, followed by a decrease by T3, although FFM values remained higher than T0. Fat-free mass gain was primarily driven by increases in the lower limbs. Fat-free mass changes between all methods were significantly correlated, with correlation coefficients of 0.70–0.97. No significant differences between time points were observed for absolute fat mass or body fat percentage, although significant correlations between several methods for change values were observed. Select laboratory and field methods can detect changes in FFM over the course of a season in elite, professional soccer athletes, with a more limited ability to detect changes in adiposity-related variables. For SKF in this population, the equation of Reilly is recommended.

Key Words: football, soccer, fat-free mass, lean mass, professional athletes, skinfolds

Introduction

Body composition estimates are commonly used in athletic settings because they can provide a means for coaches and trainers to determine athletic potential, quantify the efficacy of a training or nutrition program, or assist in preparing each athlete for an upcoming competitive season (3). Tracking changes in body composition over a competitive season can allow comprehensive athlete-monitoring assessments as well as provide information regarding each athlete's body composition adaptations during specific intervals of the playing season and offseason (8,46). While many sports contain similarities regarding favorable body composition indices to be successful, the principle of specificity still holds true in that programs should be tailored to the unique characteristics of each sport (16). Soccer is a high-intensity, intermittent field-based team sport where appropriate body fat levels enable players to move more effectively during training and games (3). However, evaluating total and regional fat-free mass (FFM) is also imperative because of its association with improving sports performance and minimizing injury risk (24).

Various approaches can be applied to examine body composition. Among them, dual-energy x-ray absorptiometry (DXA) is one of the most commonly used in athletic settings and is sometimes considered a "criterion standard" of body composition assessment (21), particularly when segmental estimates are needed. Although DXA is becoming increasingly more available, it does come with some associated concerns. Elite athletes may be reluctant to interrupt what is for many a full-time occupation for the sake of body composition assessment, thereby making the more involved laboratory techniques less appealing. Furthermore, given the cost of DXA, logistical constraints to performing a test, and the need for trained personnel, select field methods are also used to estimate body composition in athletes. Anthropometry, one of the most common field methods in assessing body composition (27), typically necessitates the use of regression equations developed in multifarious populations (23). Although skinfolds (SKF) represent an inexpensive technique requiring minimal equipment, the number of anatomical sites measured and the specific equation used to predict body composition can

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create discrepancies in results (21). While some studies have investigated the accuracy of these equations in estimating body composition in specific populations (17,52), little is known about their utility when they are used on professional soccer players (23). Furthermore, differing results may potentially be explained by the different populations in which equations were validated, as well as the reference method used (23).

Owing to the greater difficulty in conducting longitudinal research with multiple body composition assessment methods, as compared with simple cross-sectional studies, questions of the comparability of different methods for detecting body composition over time remain. In addition, while some investigations have reported the agreement of methods in general (31) or in exercise-trained populations (44), less information is available in high-level athletes (8,36). Elite athletes may exhibit anthropometric differences relative to active individuals or amateur athletes (35,38). Furthermore, the physical stress induced by professional soccer training and competitive matches across a whole season may modulate players' body composition and is a major factor contributing to their physical profile and performance (30). Indeed, body composition among soccer players is likely to change during the course of the competitive season as a result of training and competition, habitual activity, and diet (24,30,46). For instance, in 18 elite male soccer players, Devlin et al. (12) reported an increase in DXA FFM from the start of preseason (55,278 ± 5,475 g) to the start of the competitive season (56,784 ± 5,168 g; $p < 0.001$), and this increase was maintained until the end of the season. Dual-energy X-ray absorptiometry fat mass (FM) decreased over the preseason period (10,072 ± 2,493 g to 8,712 ± 1,432 g; $p < 0.001$) but increased during the latter part of the competitive season. Owen et al. (30) found decreases in FM and body fat percentage (BFP) and increases in FFM from preseason to the end of a playing season in 22 elite male European soccer players using SKF. More recently, Walker et al. (46) noted increases in DXA FFM from the preseason to the early competitive season, which was followed by a decrease in FFM back to preseason values at the end of the season in 26 Australian soccer athletes. While these investigations demonstrate the occurrence of body composition changes over a competitive season, they provide limited information regarding the comparability of different laboratory- and field-based techniques.

Questions remain regarding the utility of SKF in elite soccer athletes, and particularly, which equation is the most suitable for estimating body composition in these individuals. In addition, despite the widespread popularity of SKF in field settings, other methods based on bioelectrical impedance analysis (BIA) are also commonly used. Therefore, the purpose of this study was to quantify the changes in body composition in professional soccer players over the course of a competitive playing season and compare the ability of different assessment methods to detect changes. Specifically, the ability of DXA, multiple SKF equations, BIA, and a modified 4-compartment (4C) model to detect changes in FFM, BFP, and FM was examined. We hypothesized that laboratory methods, such as DXA and the modified 4C, would exhibit more sensitivity to detect changes in body composition as compared with SKF in general, but that specific SKF equations would perform dissimilarly because of their varying development populations and coefficients for specific body sites. It was hypothesized that the equation of Reilly et al. (34) would exhibit the best performance of the tested SKF equations because of its development in a similar population of elite male soccer players.

Methods

Experimental Approach to the Problem

The present study was an observational, 1-group, 4-time point study in elite soccer players conducted to describe body composition changes over the course of a competitive playing season and compare the ability of different assessment methods to detect body composition changes. Assessments included DXA, SKF, and BIA, with multiple body composition estimation equations used. Players completed a 4- to 5-week preseason period beginning in July and ending in early August, followed by the commencement of official matches (Figure 1). The initial assessment time point (T0) took place in the early in-season period (mid-October) after 8 official matches had been played. The next assessment (T1) took place midseason (mid-December) after 18 official matches had been played. After these matches, an 8-day winter break followed by 7 additional matches was completed before the next time point (T2) during the second part of the season (mid-February). The final time point (T3) took place at the end of the season (end of April). Accordingly, each assessment was separated from the subsequent one by approximately 2 months. Changes in body composition and the associations between changes detected by using different methods were analyzed.

Subjects

A convenience sample of 21 elite soccer players (age: 23.7 ± 4.8 years, range 18–38 years; height: 185.0 ± 5.2 cm; body mass: 80.7 ± 5.5 kg; body fat: 12.8 ± 2.2%) playing for a team competing in the Italian national second league (Serie B) championship were included in the present analysis. Each subject had to meet the following criteria: (a) >8 hours training per week, (b) being outfield players, (c) having a current professional contract with the club, (d) no history of febrile illness at the time of the initial and final assessment, and (e) no severe injury over the assessment period. Subjects performed 5 training sessions per week during the study. Written informed consent was obtained from all subjects, and the research was approved by the Ethic Committee University of Milan (approval code 1052019).

Procedures

At each time point, players were assessed using DXA, BIA, SKF, and basic anthropometric procedures (e.g., mass and height). For each testing session, subjects were required to abstain from drinking, eating, ingesting medications or dietary supplements, and exercising for at least 8 hours before reporting to the training center between 8:00 and 9:30 AM. Owing to the frequency of training sessions completed by subjects, a longer duration of abstention from exercise was not possible.

Body mass and height were measured to the nearest 0.1 kg and 0.1 cm, respectively, using a scale with a mechanical stadiometer (Seca 769, Hamburg, Germany) while subjects were barefoot and wearing a bathing suit. Total and segmental estimates of lean soft tissue (LST), fat mass, and bone mineral content were obtained by DXA on a Lunar Prodigy scanner (General Electric, Boston, MA) with enCORE software (v. 17). Fat-free mass was also obtained as the sum of LST and bone mineral content. The scanner was calibrated daily before

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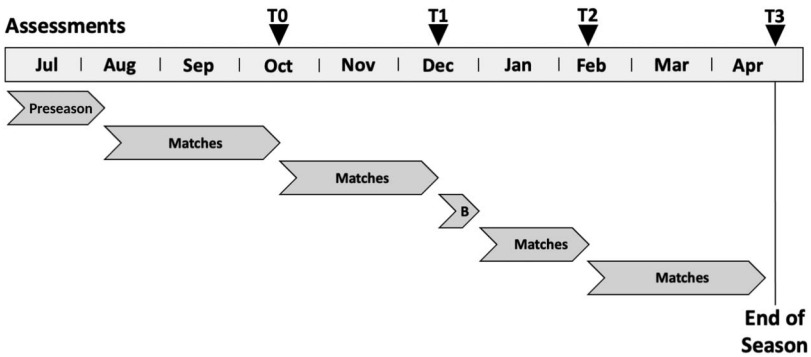

Figure 1. Study timeline. B = winter break.

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scanning according to the manufacturer's indications using a calibration block. Subjects' positioning took place using custom-made foam blocks to promote reliability and standardization of measurements, as previously indicated by Nana et al. (29). Using the enCORE software, a trained physician manually adjusted the lines used to establish boundaries between body regions. For arms, the line was placed directly through the glenohumeral joint. All tissue distal to this line, for example, from the glenohumeral joint to the tips of the phalanges, was designed as the arm segment. For legs, the line was placed through the femoral neck, directly perpendicular to the axis, and angled such that the entirety of the pelvis remained in the trunk region. All tissue distal to this line, for example, from the femoral neck to the tips of the phalanges, was identified as the leg region (5).

Bioelectrical impedance analysis was performed using a phase-sensitive bioelectrical analyzer (BIA 101 BIVA PRO, Akern, Florence, Italy). The device emits an alternating sinusoidal electric current of 250 μA at a single frequency of 50 kHz (±0.1%). The device was calibrated using the standard control circuit supplied by a manufacturer that has a known impedance. Subjects were positioned supine with a leg opening of 45° with respect to the midline of the body and with the upper limbs positioned 30° away from the trunk. After cleaning the skin with alcohol pads, 4 adhesive electrodes (Biatrodes Akern Srl, Florence, Italy) were placed on the back of the hands and 4 other electrodes on the neck of the corresponding feet, keeping 5 cm between each electrode. The proximal hand electrode was positioned between the radial and ulnar styloid processes, directly superficial to the distal radioulnar joint. The distal hand electrode was positioned in the center of the third proximal phalanx. The proximal foot electrode was placed directly between the medial and lateral malleoli at the ankle. The distal foot electrode was placed immediately proximal to the second and third metatarsophalangeal joints (7,9). Multi-frequency BIA was performed using an InBody 770 analyzer (InBody Co., Ltd., South Korea), which has been demonstrated to detect longitudinal bioimpedance changes in active individuals (40).

All anthropometric measurements were performed by an accredited level 2 anthropometrist (T.B.) following the International Society Advancement Kinanthropometry guidelines. Body mass index was calculated as the ratio of body mass to height squared ($kg \cdot m^{-2}$). Girths were measured to the nearest 0.1 cm with an anthropometric tape (Lufkin executive thinline, W606ME). Skinfold thicknesses were measured to the nearest 0.1 mm with a skinfold caliper (Holtain Ltd, Crymych, United Kingdom). For each anthropometric point considered, 3 nonconsecutive measurements were performed to compute the average. The technical error of measurement scores was required to be within 5% agreement for SKF and within 1% for breadths and girths (6).

Calculations. Select skinfold equations were chosen by the researchers who also served as nutrition and sport science personnel for the professional soccer team being examined. These included the equations of Civar et al. (10,23), Faulkner et al. (13,15), Reilly et al. (34), Yuhasz (51), Withers et al. & Siri (37,50), and Wilmore et al. & Siri (37,48). In addition, the FFM prediction equation of Matias et al. (25), which includes bioelectrical resistance, height, weight, and sex, was used to generate FFM, FM, and BFP estimates. Bioelectrical resistance was obtained from a supine BIA analyzer (Akern, BIA 101 BIVA PRO, Florence, Italy).

A modified 4C model was produced using DXA, bioimpedance, and scale data. Body volume (BV) was estimated from DXA data using the equation of Wilson et al. (49):

$$BV\,(L) = 0.933*LST + 1.150*FM + -0.438*BMC + 1.504$$

where BMC is bone mineral content. Total body water (TBW) was estimated from bioimpedance and anthropometric data using the equation of Matias et al. (26):

$$TBW\,(L) = 0.286 + 0.195*S2/R + 0.385*BM + 5.086*sex$$

Subsequently, 4C FM was estimated using the equation of Wang et al. (47):

$$FM\,(kg) = 2.748*BV - 0.699*TBW + 1.129*Mo - 2.051*BM$$

4C FFM and BFP were then calculated.

Statistical Analyses

An a priori statistical power analysis was conducted to determine the sample size using G*Power statistical software (14). Consequently, given the study design and an effect size falling between "small" and "medium" (Cohen's $f = 0.175$), the estimated required sample size was 20 subjects.

Data were analyzed in R (32) (v. 4.1.2) using the *rstatix* software package (22) (v. 0.7.0). To evaluate how different methods quantified body composition changes over time, 2-way repeated measures analysis of variance (ANOVA) tests were performed for FFM, FM, and BFP, with method and time as within-subjects factors. When a statistically significant

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method × time interaction was observed, follow-up testing was performed using one-way ANOVAs for simple main effects of method and time, as well as subsequent pairwise comparisons using *t*-tests, with correction for multiple comparisons using the Benjamini and Hochberg adjustment (2). To evaluate changes in segmental body composition and anthropometric variables, one-way repeated measures ANOVA tests were performed, with time as a within-subjects factor. When a statistically significant effect of time was observed, follow-up pairwise comparisons were performed using *t*-tests, with correction for multiple comparisons using the Benjamini and Hochberg adjustment (2). Data were examined for normality and outliers using Shapiro-Wilk tests, inspection of quantile-quantile plots, and boxplot methods. When violations of the sphericity assumption were detected using Mauchly's test, the Greenhouse-Geisser correction was applied. For these tests, statistical significance was set at $p < 0.05$.

Pearson's correlations between FFM and FM changes detected by DXA and each other method were examined for each pair of adjacent time points (i.e., changes from T0 to T1, T1 to T2, and T2 to T3). For these tests, statistical significance was set at $p \leq 0.0019$ (0.05 divided by 27 tests per body composition variable). For these comparisons, DXA rather than 4C was used due to the greater availability and frequency of use, as well as the necessity to use DXA in subsequent correlations using segmental changes.

Pearson's correlations between total and segmental changes in DXA FM and changes in total and segmental skinfold thicknesses were examined for each pair of adjacent time points (i.e., changes from T0 to T1, T1 to T2, and T2 to T3) (18). For correlations involving total FM changes, statistical significance was set at $p \leq 0.0019$ (0.05 divided by 27 tests). For correlations involving segmental FM changes, statistical significance was set at $p \leq 0.0021$ (0.05 divided by 24 tests).

Results

Comparison of Methods (Total Body Composition)

Statistically significant condition × time interactions were observed for FFM, FM, and BFP ($p \leq 0.0005$ for each; Figure 2). Follow-up one-way ANOVAs indicated simple main effects of time were present for FFM in all methods ($p \leq 0.05$ for each); however, no method demonstrated a simple main effect of time for FM or BFP ($p = 0.06$ to $p = 0.96$). Follow-up pairwise comparisons for FFM demonstrated significant differences between time points for only 4 methods (DXA, 4C model, Civar equation, and Reilly equation), with the largest number of distinct changes detected using DXA. For DXA, FFM values differed between T0 and each other time point, as well as between T1 and T2 and between T2 and T3 (Figure 2A). For 4C, FFM differed between T0 and T2 and T3, as well as between T1 and T2 and between T2 and T3. For the Civar and Reilly equations, significant differences between T0 and T2 and between T1 and T2 were detected.

Follow-up one-way ANOVAs indicated simple main effects of method at all time points for FFM, FM, and BFP ($p < 0.0001$ for all). Dual-energy x-ray absorptiometry differed from each other method at every time point ($p < 0.05$ for all). 4C differed from each other method at every time point ($p < 0.05$ for all), except for InBody and Withers at T0 and T1 and Yuhasz at all 4 time points ($p > 0.05$ for all). Inbody differed from each other method at every time point ($p < 0.05$ for all), except for Reilly, Wilmore, and Withers at all time points and 4C at T0 and T1 ($p > 0.05$ for all). Absolute and change values for each time point are displayed in Supplemental Digital Content 1 (see Table 1, http://links.lww.com/JSCR/A440).

For FFM changes, statistically significant correlations between DXA and each other method were observed for each time interval (i.e., T0 to T1, T1 to T2, and T2 to T3; Figure 3 and see Figures, Supplemental Digital Content 2, http://links.lww.com/JSCR/A441). Pearson correlation coefficients (r) for FFM changes between DXA and each other method ranged from 0.70 to 0.97. For T0 to T1, statistically significant correlations between FM changes detected by DXA and each other method were observed (see Figure, Supplemental Digital Content 3, http://links.lww.com/JSCR/A442), except for the Matias equation ($p = 0.024$) because of p-value correction for multiple comparisons. For T1 to T2, statistically significant correlations between FM changes detected by DXA and each other method were observed, except for Inbody ($p = 0.01$) and the Matias equation ($p = 0.04$) because of p-value correction. For T2 to T3, statistically significant correlations between FM changes detected by DXA and each other method were observed, except for the Civar equation, Inbody, and the Matias equation. Overall, correlation coefficients for FM changes between DXA and each other method ranged from 0.25 to 0.90.

For T0 to T1, statistically significant correlations between BFP changes detected by DXA and each other method were observed, except for Inbody, the Matias equation, and the Wilmore equation (see Figure, Supplemental Digital Content 4, http://links.lww.com/JSCR/A443). For T1 to T2, statistically significant correlations between BFP changes detected by DXA and each other method were observed, except for Inbody and the Matias equation. For T2 to T3, statistically significant correlations between BFP changes detected by DXA and 3C, the Faulkner equation, and the Wilmore equation were observed. Overall, correlation coefficients for BFP changes between DXA and each other method ranged from 0.14 to 0.88.

Segmental and Anthropometric Changes

Changes in LST of the arms and legs were observed at specific time points (Figure 4). Arms LST decreased from T0 to T1, increased to baseline (T0) values at T2, and subsequently decreased again from T2 to T3 (Figure 4B). Legs LST increased from T0 to T2 only, an effect that was also observed in both the right and left legs individually (Figure 4D–F). No changes in trunk LST were detected (Figure 4C). No changes in segmental FM were observed (Figure 5), except for an increase in right leg FM from T2 to T3 (Figure 5E).

A change in TBW over time was detected, with values increasing from T1 to T2 (see Figures, Supplemental Digital Content 5, http://links.lww.com/JSCR/A444). Although a significant effect of time was also detected for intracellular water, follow-up pairwise comparisons did not reveal statistically significant differences between any time points. No changes in extracellular water were detected. Total BMC increased from T0 to T2 and T3 (see Figure, Supplemental Digital Content 4, http://links.lww.com/JSCR/A443). BMC of the arms decreased from T0 to T1 and from T2 to T3, with the T3 value also being lower than T0. BMC of the left leg and both legs combined was higher at T1, T2, and T3 as compared with T0. However, no significant changes in BMD were observed.

No changes in the sum of 8 skinfold thicknesses were observed (Figure 6A). However, triceps skinfold thickness decreased from T1 to T3 (Figure 6B), while thigh skinfold thickness increased

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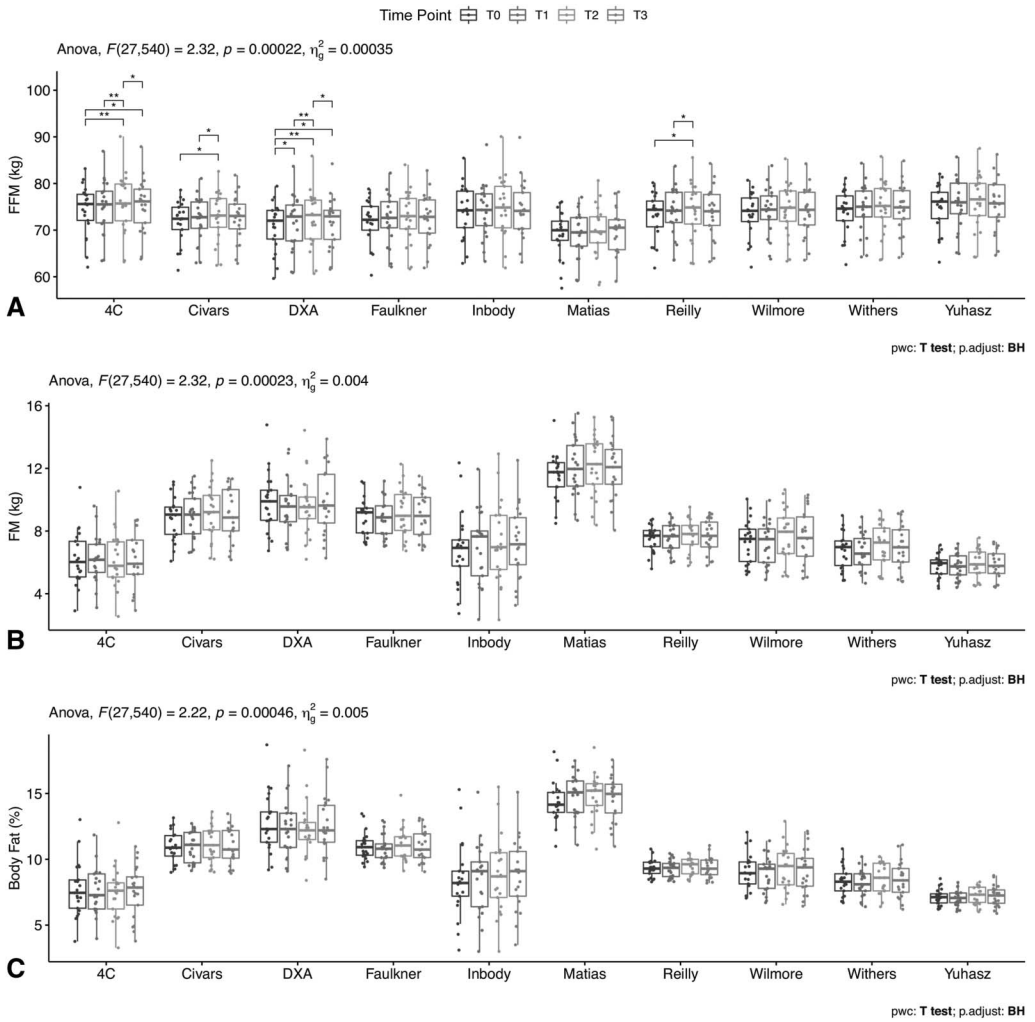

Figure 2. Total body composition changes. Changes in total fat-free mass (FFM, panel A), fat mass (FM, panel B), and body fat (%) (panel C) detected by using body composition assessment methods over the course of a competitive season in professional soccer athletes.

from T1 to T2 and T3 (Figure 6H). Most circumferences did not change over the measurement period (see Figures, Supplemental Digital Content 5, http://links.lww.com/JSCR/A444). However, hip circumference decreased from T2 to T3. No significant changes in bioimpedance variables were observed (see Figures, Supplemental Digital Content 5, http://links.lww.com/JSCR/A444).

Correlations Between DXA Fat Mass and Skinfold Thicknesses

Correlations between DXA and raw SKF are displayed in Supplemental Digital Content 6 (see Figures, http://links.lww.com/JSCR/A445). Between T0 and T1, DXA total FM changes were correlated with changes in the abdominal skinfold site ($r = 0.80$, $p < 0.001$), as well as the sum of 8 SKF ($r = 0.67$, $p < 0.001$). Between T1 and T2, DXA total FM changes were correlated with changes in the abdominal ($r = 0.88$, $p < 0.001$) and subscapular ($r = 0.70$, $p < 0.001$) skinfold sites, as well as the sum of 8 SKF ($r = 0.84$, $p < 0.001$). Between T2 and T3, DXA total FM changes were correlated with changes in the abdominal ($r = 0.66$, $p = 0.001$), iliac crest ($r = 0.68$, $p < 0.001$), and subscapular sites ($r = 0.70$, $p < 0.001$), as well as the sum of 8 SKF ($r = 0.72$, $p < 0.001$). Changes in skinfold thickness values of the biceps and triceps were not correlated with changes in DXA Arms FM at any time interval. Similarly, changes in skinfold thickness values of the calf and thigh were not correlated with changes in DXA Legs FM at any time interval. However, changes in abdominal skinfold thickness were correlated with changes in DXA trunk FM at each time interval (i.e., T0 to T1, T1 to T2, and T2 to T3; range of r: 0.70 to 0.83, $p < 0.001$ for all). In addition, changes in iliac crest skinfold thickness from T2 to T3 were correlated with DXA trunk FM changes ($r = 0.65$, $p = 0.0015$).

Discussion

The purpose of this study was to describe body composition changes in professional soccer players over the course of a competitive playing season, compare the ability of different assessment methods to detect changes over the course of a season, and examine the relationship between segmental DXA- and skinfold-derived variables to help inform sports science practitioners working in limited-resource settings. The 4 time points examined spanned from the early in-season period (mid-October) to the end of the competitive season (i.e., the end of

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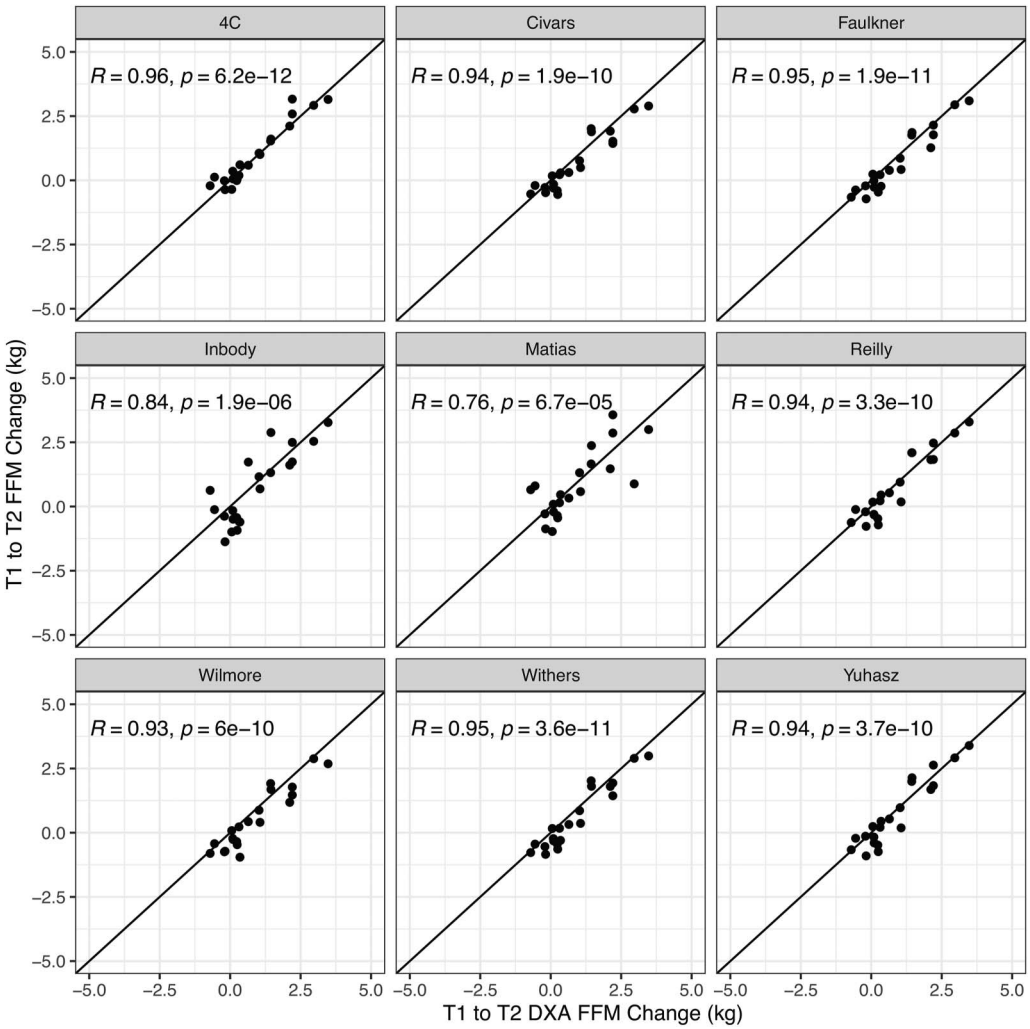

Figure 3. Correlations between methods for fat-free mass changes from T1 to T2. Pearson's correlations for changes in fat-free mass (FFM) between dual-energy x-ray absorptiometry (DXA) and other methods during the mid-December (T1) to mid-February (T2) interval in professional soccer athletes.

the following April). With the most advanced methods in this study, DXA and the modified 4C model, statistically significant changes in FFM were detectable between most time points. Specifically, DXA detected significant differences for 5 of 6 possible comparisons between pairs of time points, while the 4C model detected significant FFM differences in 4 of 6 possible comparisons. Generally, both methods detected increases in FFM from T0 (mid-October) up to peak levels at T2 (mid-February), which were ~1.6 kg higher than T0 on average. This was followed by a ~0.75 kg decrease from T2 to T3 (mid-April), although FFM remained above initial T0 values. Two SKF equations, the equations of Reilly et al. (34) and Civar et al. (10,23), also detected significant FFM increases between 2 pairs of assessments. For both equations, these increases were observed for the four-month in-season period from T0 (mid-October) to T2 (mid-February) and the two-month in-season period from T1 (mid-December) to T2 (mid-February). In addition, the magnitude of increase for the Reilly equation (~1.2 kg from T0 to T2) and Civar equation (~1.1 kg) was similar, although slightly smaller magnitude than DXA changes (~1.6 kg). No other SKF methods or BIA methods detected changes between time points. Collectively, these results indicate that levels of FFM increased from the early in-season period up to a peak approximately 2 months before the end of the competitive season, with these changes detectable by both laboratory methods and select SKF equations. Segmental DXA analysis further revealed that the increase in FFM was driven by the significant increases in LST of both legs. In contrast to the FFM results, no significant FM or BFP changes between time points were observed for any method. These findings demonstrate that laboratory-grade and select SKF-based methods are better able to detect statistically significant changes in FFM as compared with adiposity-related variables.

Previous research has examined body composition changes in soccer players over the course of a playing season, with the inclusion of multiple assessment methods. In a sample of 26 Australian soccer athletes, Walker et al. (46) reported increases in DXA FFM from the preseason to the early competitive season, followed by a decrease in FFM back to preseason values at the end of the season. While the different time points prevent an optimal comparison, the present finding of

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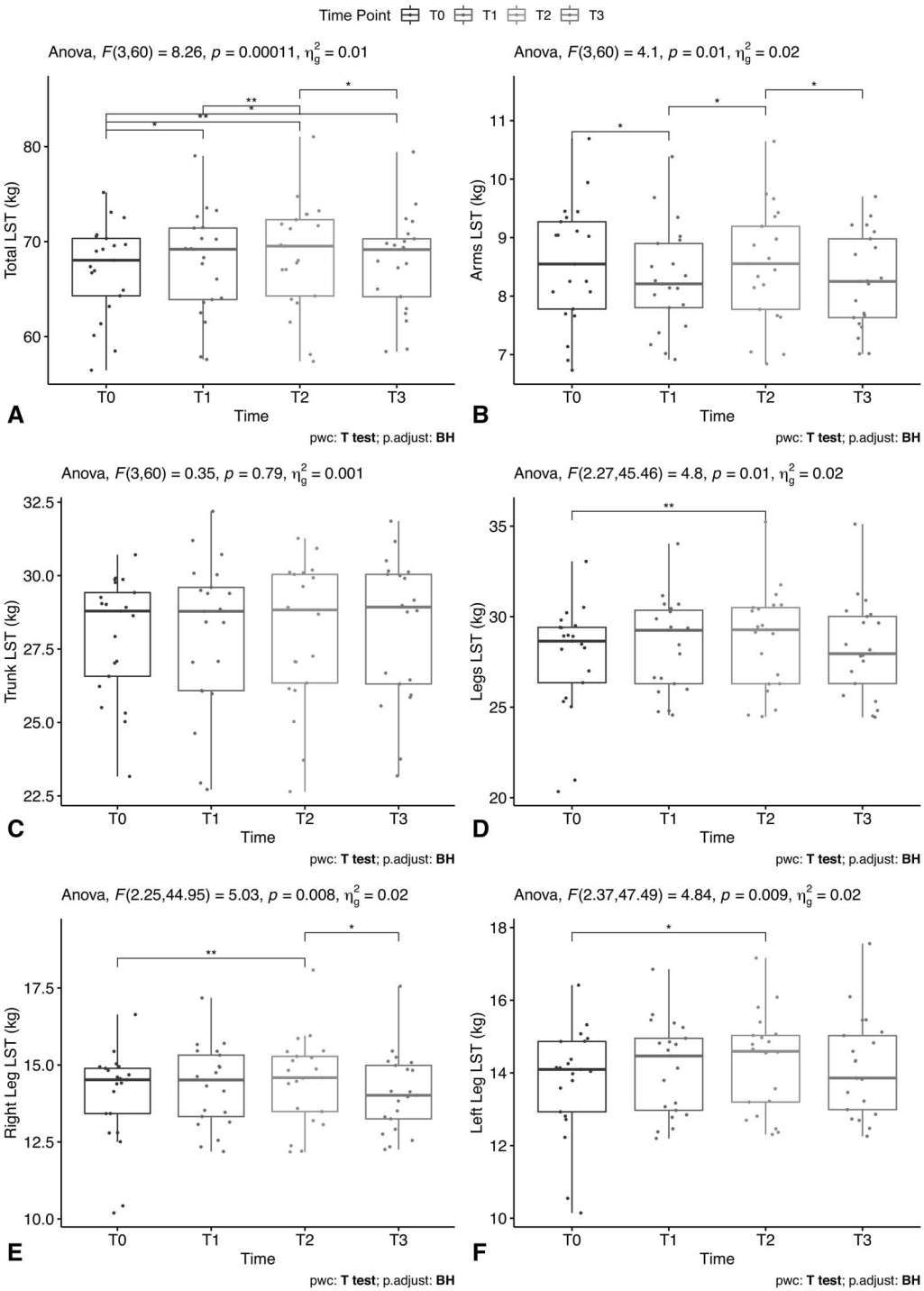

Figure 4. Segmental lean soft tissue (LST) changes from dual-energy x-ray absorptiometry. Changes in total (A) and segmental LST over the course of a competitive season in professional soccer athletes. Segmental changes include the arms (B), trunk (C), legs (D), right leg (E), and left leg (F).

decreases in FFM during the last ~2 months of the playing season may agree with these previous findings. However, in the present investigation, FFM values remained ~0.75 kg above early in-season levels at the end of the playing season. In the same previous investigation (46), seven-site skinfold thickness decreased over the season. While this was not observed for raw skinfold thicknesses in this study, 2 of the skinfold-based equations did detect changes in FFM. Other research has indicated a relative lack of body composition changes in soccer athletes over 3 years of playing (39), although this work was conducted in female collegiate soccer players rather than professional male soccer players as in the present study. Body composition and overall body phenotype vary across competitive levels (38), with some body composition reference data available for professional soccer players. Relative to professional Mexican soccer players from multiple divisions

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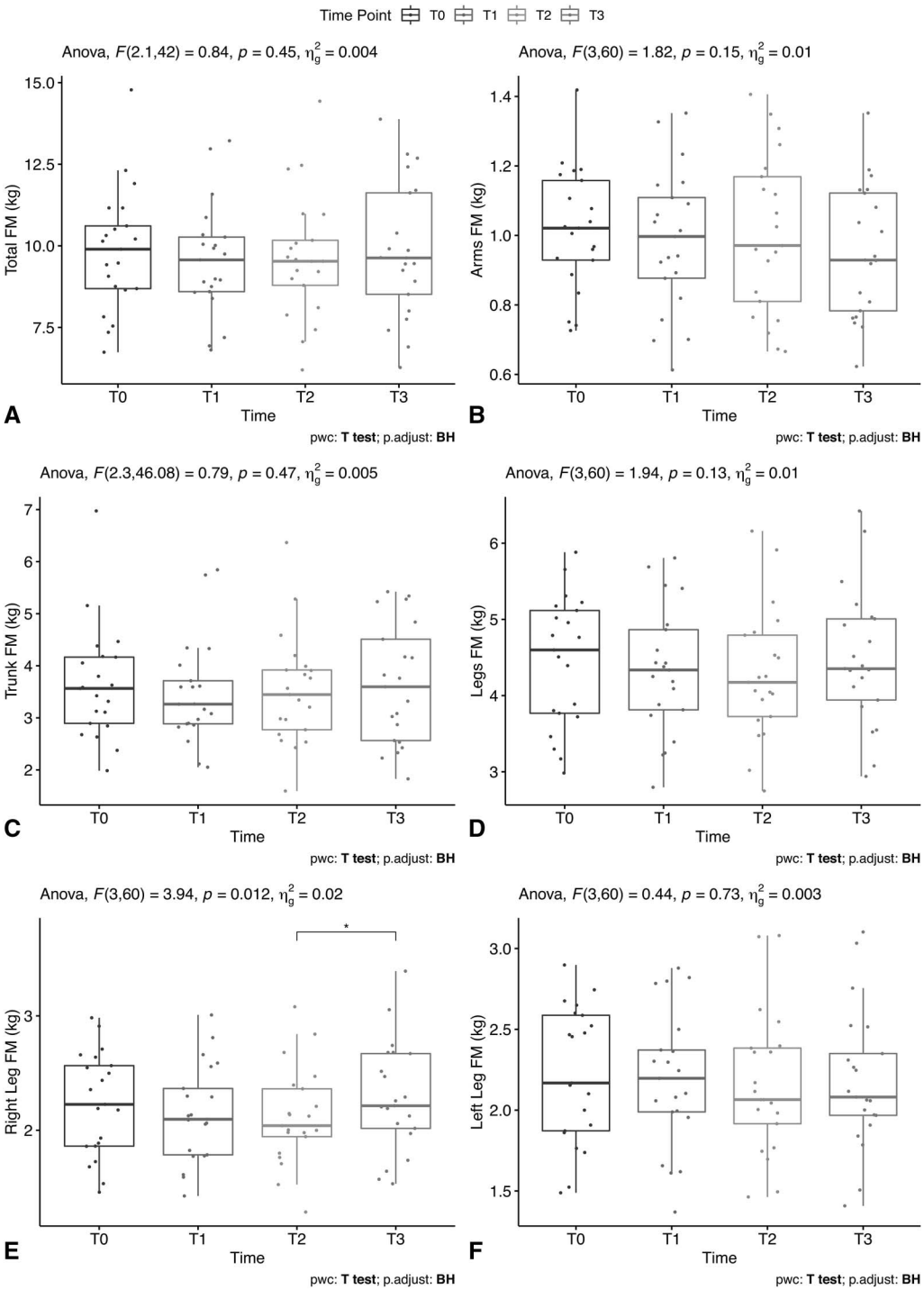

Figure 5. Segmental fat mass changes from dual-energy x-ray absorptiometry. Changes in total (A) and segmental fat mass (FM) over the course of a competitive season in professional soccer athletes. Segmental changes include the arms (B), trunk (C), legs (D), right leg (E), and left leg (F).

(fourth division, third division, U-17, second division, and U-20) (3), the present sample of professional Italian athletes was older and taller and displayed higher body mass but lower relative body fat. By contrast, our subject characteristics were more similar to those of English Premier League players described in previous research (41). Collectively, expected body composition characteristics and changes over a season in soccer players may vary based on the competition level, training status, demographic characteristics, and timing of assessments, in addition to the sensitivity of the assessment method itself.

Regarding absolute body composition values in this study, DXA differed from each other method at every time point, with the modified 4C model also differing from nearly every other method at each time point. Multifrequency BIA values were also different from most methods at most time points. Despite

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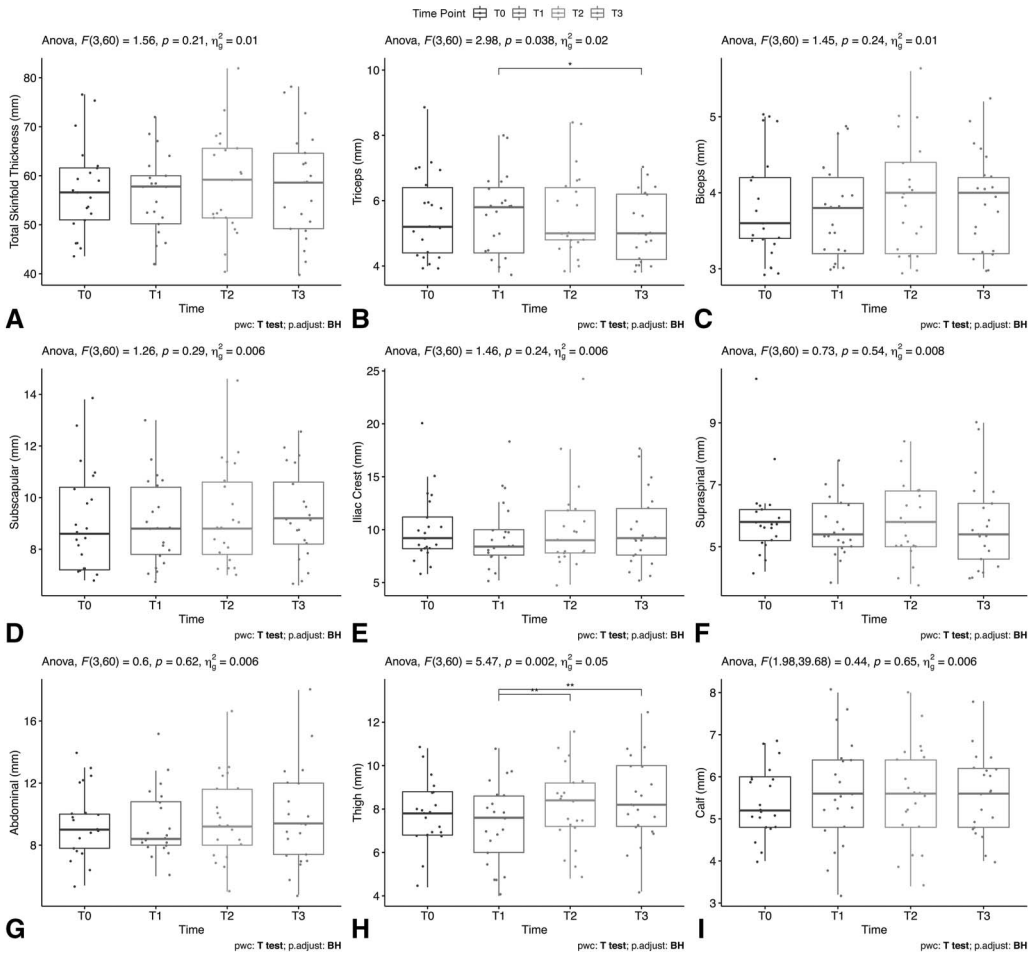

Figure 6. Skinfold thickness changes. Changes in raw skinfold thicknesses over the course of a competitive season in professional soccer athletes. Changes are displayed for total skinfold thickness (A) and the skinfold thicknesses of the triceps (B), biceps (C), subscapular (D), iliac crest (E), supraspinal (F), abdominal (G), thigh (H), and calf (I).

these differences in absolute values, statistically significant correlations between DXA FFM changes and FFM changes quantified by using each other method were observed for each set of time points examined (i.e., T0 to T1, T1 to T2, and T2 to T3), with correlation coefficients ranging from 0.70 to 0.97. In addition, despite the apparent lack of ability to detect statistically significant changes in absolute FM values, DXA FM change values were correlated with changes detected by using most other methods for each set of time points, although correlations ranged widely from 0.25 to 0.90. Similar to FM, DXA BFP change values were correlated with most other methods, but correlation coefficients ranged from negligible (0.14) to strong (0.88). Nonetheless, these correlational analyses indicate stronger between-method agreement for FFM changes as opposed to adiposity-related variables. This finding—along with the previous discussion regarding the detection of absolute FFM changes by DXA, the modified 4C model, and select SKF equations, as well as the functional importance of FFM (42)—may indicate that FFM is a preferred total body composition metric for tracking during a competitive season. Previous research has also demonstrated lower relative error (i.e., better reliability) of FFM estimates across a variety of body composition assessment methods, as compared with FM, largely because of higher absolute FFM values and small absolute FM values, particularly in lean athletes (44).

In addition to a consideration of total body composition changes, alterations in specific body segments over a playing season were examined because of their potential importance in soccer athletes. Some previous research has indicated the presence of between-leg differences in muscle size and performance in soccer athletes, likely because of the varying muscular requirements of the dominant vs. nondominant limb (4,33). However, other investigations have reported a relative lack of muscular asymmetries, potentially because of the longitudinal effects of a playing career or functional compensation by players (11), or even an attenuation of the muscular imbalances seen in nonathletes (19). While, as expected, total DXA LST demonstrated the same significant differences between time points as DXA FFM (i.e., LST + BMC), segmental differences were less consistent. Arms LST differed for 3 of 6 possible comparisons, while total leg LST and left leg LST differed for only one comparison, and right leg LST differed for 2 comparisons. As mentioned, the increase in LST of both legs seems to be the factor responsible for the elevation in total FFM and LST. Overall, the segmental LST findings indicate that while DXA possesses the ability to detect segmental LST changes during a competitive season, particularly for the lower limbs, small differences at select time points may be more difficult to detect as compared with changes in whole-body values. This agrees with previous research exploring total body

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composition assessment as compared with segmental values (28,45). For segmental FM, the sole significant change was an increase in right leg FM from T2 (mid-February) to T3 (the end of April and end of the playing season).

Beyond DXA, skinfold measures also allowed for a consideration of segmental changes of specific subcutaneous fat depots. Skinfolds are an attractive body composition assessment option in field settings because of their low cost, convenience, and the multiple values they provide (21). Indeed, a survey conducted by an ad hoc research group under the auspices of the International Olympic Committee found that SKF were the most used body composition assessment method among 216 responding organizations from 33 countries (27). Of these respondents, 50% used International Society for Kinanthropometry (ISAK) SKF procedures, 40% used non-ISAK procedures, and 28% used both. Interestingly, in this study, significant differences between raw skinfold thicknesses were seen only for the triceps (T1 vs. T3) and thigh (T1 vs. T2 and T1 vs. T3). However, as noted, 2 skinfold-based equations using the raw skinfold values detected changes in FFM at some time points.

The finding that the skinfold equation of Reilly et al. (34) was able to detect select changes in FFM is noteworthy because, unlike other equations, it was developed specifically for professional soccer players. This may indicate the population-specific nature of skinfold-based equations. In addition, the Reilly equation included thicknesses of the triceps, abdomen, thigh, and calf SKF. As noted, only the triceps and thigh SKF demonstrated statistically significant changes between time points. Significant changes in both raw thigh skinfold thickness and Reilly FFM were observed from T1 to T2, although the other significant change in thigh skinfold thickness (T1–T3) and the significant change in triceps thickness (T1–T3) did not correspond to a significant difference in FFM with the Reilly equation. The other SKF-based equation to detect significant changes in FFM, the Civar equation (10,23)—which included triceps, abdomen, and biceps SKF and body mass—was not developed in soccer players; however, all subjects were active male subjects involved in either team (basketball, soccer, handball, and volleyball) or individual sports (track and field, tennis, and combat). As such, these subjects could have exhibited greater similarity with the present sample than the development groups for some other equations for general exercise training status. However, based on the greater similarity of the development population, the Reilly equation may still be viewed as the preferred SKF equation for elite male soccer athletes.

While multicompartment models are typically viewed as criterion methods for total body composition assessment at the molecular level (43), the greater availability of DXA and ability of DXA to produce segmental variables have led to its common use as a reference method of body composition assessment in sport settings (1,28). In the present analysis, the use of DXA variables also allowed a consideration of the relationship between raw skinfold thickness changes and corresponding FM changes in the DXA regions containing those skinfold sites. Previous research demonstrated a Pearson's correlation of 0.49 for percent changes in ultrasound-derived subcutaneous thickness (sum of 7 sites specified by Jackson and Pollock (20)) and DXA FM in response to a combined resistance training program and hypercaloric diet in resistance-trained male subjects (18). In this study, higher correlations were observed for changes in DXA FM and the sum of 8 SKF, with Pearson's correlation coefficients of 0.67–0.84 depending on the time points being considered. These findings, along with additional analyses of correlations in individual body regions, may aid sport scientists and practitioners seeking to better understand the relationship between laboratory and field methods of body composition assessment.

Strengths of this study include the professional athletes who served as subjects, the multiple time points and assessment methods used, and the rigorous standardization before assessments, which is often difficult to implement in elite athletes during the competitive season. Another unique aspect of this study was the multifaceted approach that considered relatively unexplored relationships, such as the association between changes in specific SKF sites and segmental body composition estimates from DXA, as well as the performance of multiple field-based methods as compared with laboratory methods like DXA. Limitations include the absence of a true preseason assessment, with the first time point occurring after 8 official matches were played, and the inclusion of only male athletes.

Practical Applications

In elite, professional male soccer players, FFM and LST may increase during the playing season, up to a peak approximately 2 months before the end of the season. This increase may be primarily driven by gains in the lower limb. While a small decrease in FFM and LST was detected during the final 2 months of the playing season, values at the end of the season remained above early in-season values. Knowledge of these changes may help inform strength and conditioning personnel, sports medicine staff, coaches, and soccer athletes of expected variation in body composition during a competitive playing season. Although laboratory-based body composition methods (such as DXA and a modified 4C model) can detect changes in FFM, these techniques are often inaccessible to practitioners. As such, SKF may represent a practical alternative. When using SKF, the equation of Reilly et al. (34) is recommended for use in elite male soccer players. This equation was developed in a relevant group of professional soccer players and demonstrated a strong correlation with DXA for FFM changes in this study, as well as a relatively similar magnitude of average FFM change as compared with DXA. In addition, the equation of Civar et al. (10,23) demonstrated comparable performance with the Reilly equation, although it was developed in players from a variety of team and individual sports. Nonetheless, practitioners should be aware that these field methods may exhibit less sensitivity than laboratory methods, such as DXA, indicating that a larger change in FFM may potentially be needed to confidently detect a change. Using skinfold thicknesses measured in mm, FFM can be estimated from the Reilly et al. equation (34) as

$$\text{FFM} = \text{Body mass} - \text{Body mass}^{*}\frac{(5.174 + (0.124 \times \text{thigh}) + (0.147 \times \text{abdomen}) + (0.196 \times \text{triceps}) + (0.130 \times \text{calf}))}{100}$$

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